

LUBRICATION

Section 6

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LUBRICATION

Lubrication is one of the most important items in the proper operation and maintenance of the Automatic Pinspotter.

Care must be taken to insure correct lubrication. Avoid excessive use of lubricants to prevent the possibility of transmitting it to the bowler.

The following pages indicate the points of lubrication, the correct lubricant and the frequency of lubrication for each part of the machines.

Before lubricating exposed parts or surfaces, it is important that the old lubricant first be removed.

LUBRICANT SYMBOLS



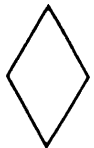
Items marked with this designation pertain to oiling. The lubricant should be an SAE #10 oil. The location, amount, and frequency of oiling is noted on each individual page.



Items marked with this designation pertain to greasing. The lubricant should be multi-purpose grease (Bearing Gard #2). The location and frequency of greasing is noted on each individual page.



Items marked with this designation pertain to the lubrication of the gear head motors used on table, sweep, and back end. The following page provides the recommended lubricants. The individual pages will provide the location and frequency to use.



The item marked with this designation applies to the recommended lubrication of the pin elevator wheel ring tube only. The following page provides the recommended lubricants. The individual pages will provide the location and frequency of use.

RECOMMENDED LUBRICATION OVERALL MACHINE LUBRICATION

AMF recommends the use of an SAE #10 motor oil for overall machine lubrication. This is obtainable locally at service stations or automobile supply stores.

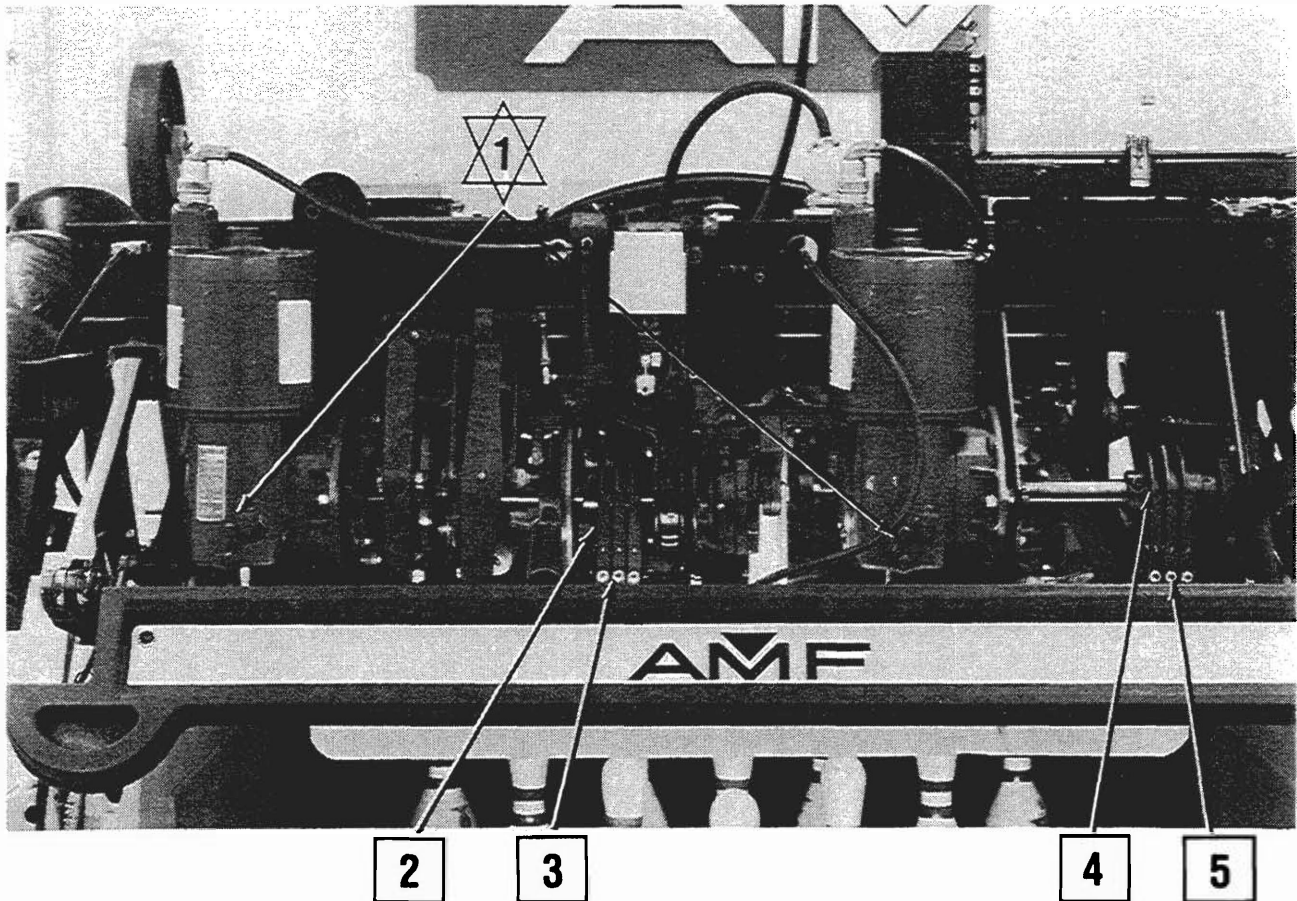
A special grease containing lithium is to be used for overall machine lubrication including the rod end bearings. Bearing Gard #2, a product of the Standard Oil Company of Ohio, is recommended and if not obtainable locally, may be ordered from AMF under part number 715 010 307 for a one pound tube or part number 715 010 407 for a one pound can. This grease is recommended for use on ball, needle, roller and rod end bearings.

A special oil known commercially as Way Oil, SAE #80, is recommended for lubricating the ring tube supporting the pin elevator wheel. The Way Oil is wick fed to the bearing surface of the ring tube and will adhere to the surface without dripping. If this oil is not obtainable locally it may be ordered from AMF under part number 715 021 006.

GEAR MOTOR LUBRICANTS

The table, sweep and back end gear motors are factory filled with J-156 oil, (HLP 680), a product of the Standard Oil Company of Ohio. This oil is also available from AMF under part number 715 020 906.

CAUTION: Do **not** use or add any other oil. Oils of different grades or different manufacturers are not compatible to one another and will not provide the desired 5-year service life of J-156 oil (HLP 680).



FRONT END ASSEMBLY



1. Check oil level of table and sweep motors once per year by removing oil level plug. Fill as required with J-156 (AMF Part No. 715 020 906).

2. Drain and refill once every five years. If oil has sludged or contamination, drain, flush with kerosene, rinse with J-156 and refill to level plug with J-156 (AMF part no. 715 020 906).



Table cam lever rollers (3 places) 1 drop #10 oil every 3 months.



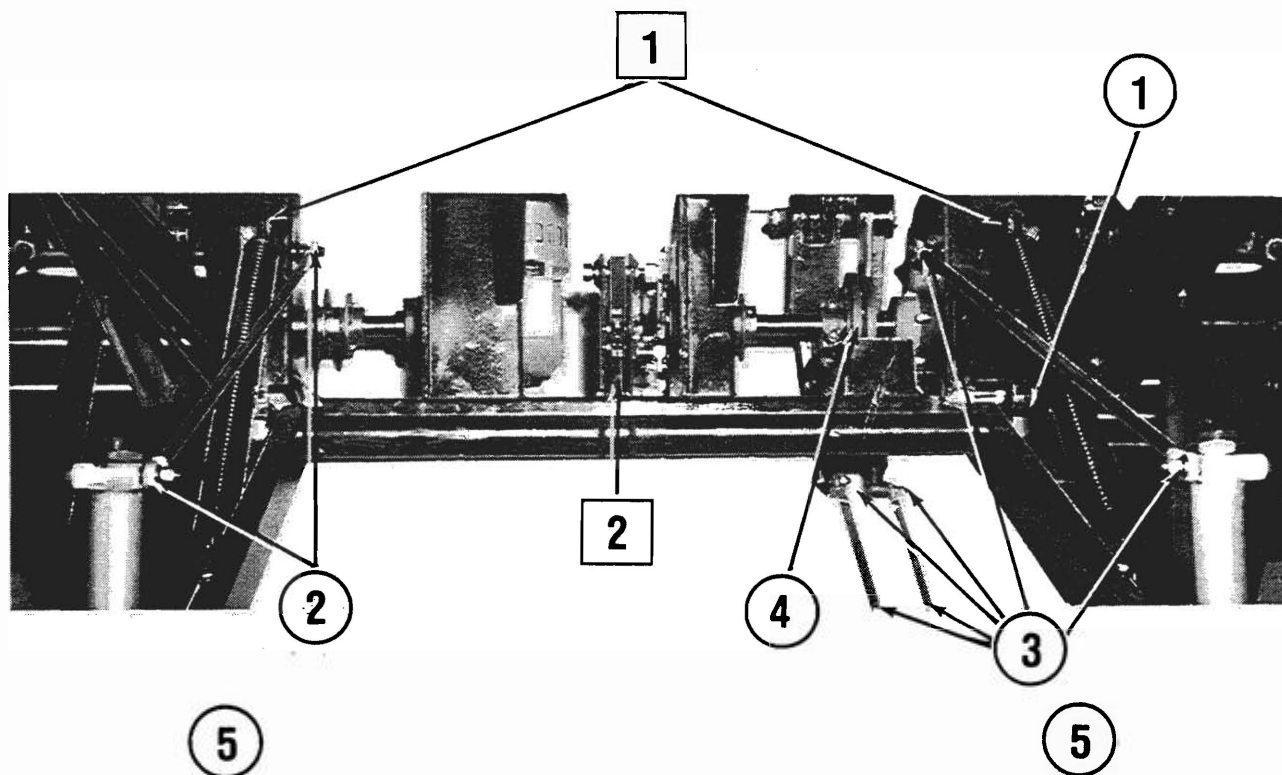
Table cam lever pivots (6 places) 1 drop #10 oil every 3 months.



Sweep cam lever rollers (3 places) 1 drop #10 oil every 3 months.

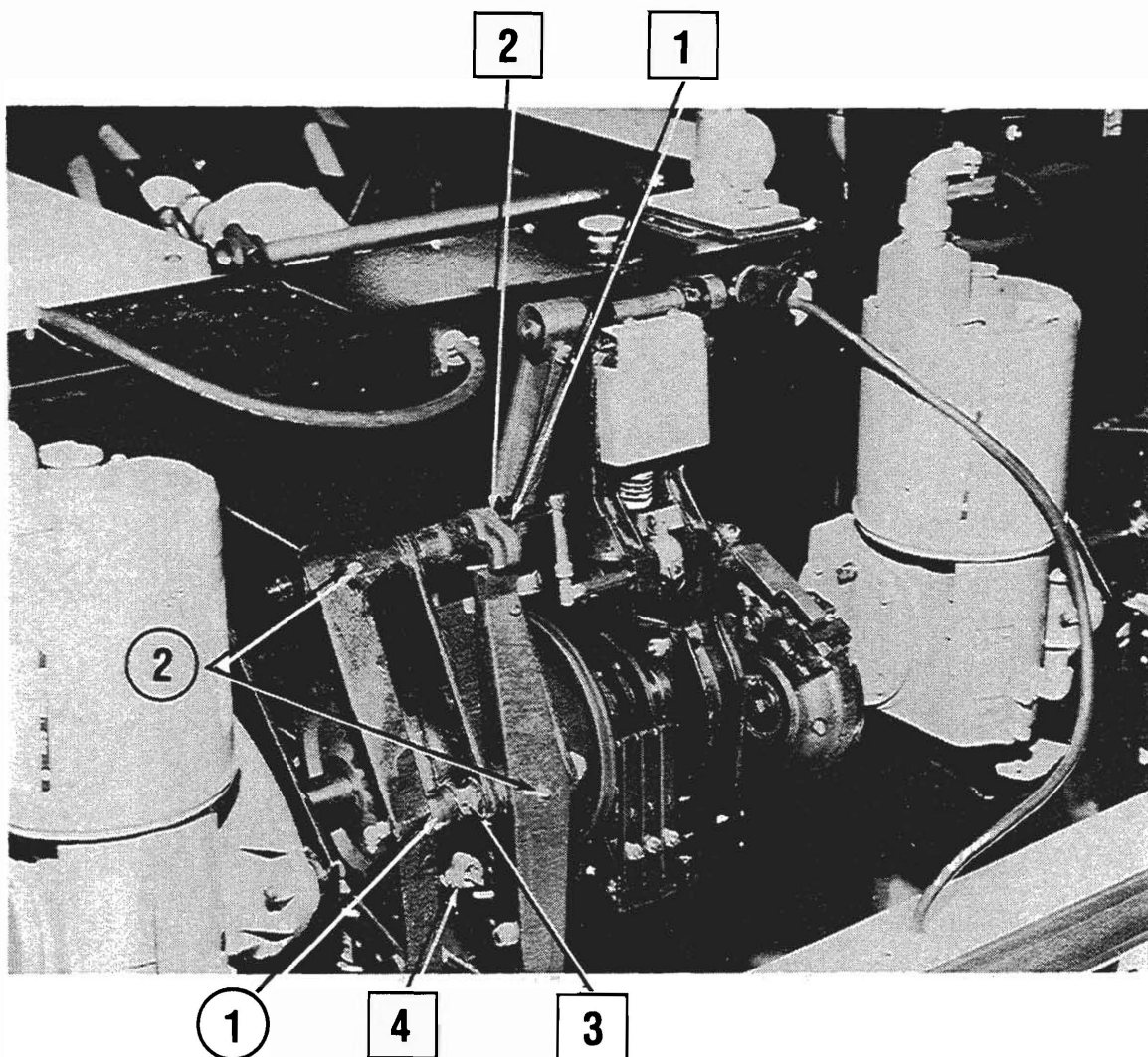


Sweep cam lever pivots (6 places) 1 drop #10 oil every 3 months.



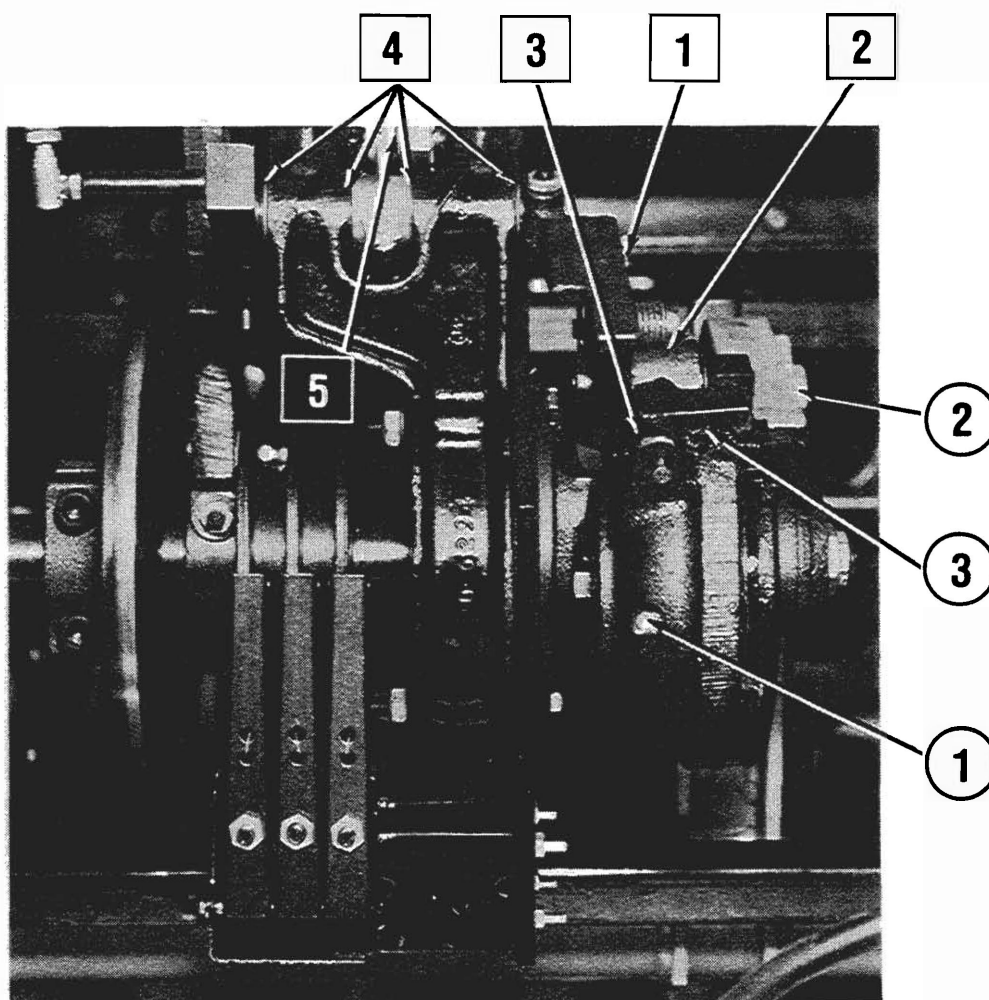
FRONT END ASSEMBLY

- 1** Spring roller bearing (2 places) 2 drops #10 oil every 3 months.
- 2** Over travel lever pivot (2 places) 1 drop #10 oil every 6 months.
- 1** Pillow blocks on torque tube weldment (2 places) every 3 months.
- 2** Table connecting rod ends (2 places) grease every 3 months.
- 3** Connecting rod ends (6 places) every 3 months.
- 4** Table spot and respot cams (2 places) apply light coating grease every month.
- 5** Spherical bushing in table support weldment (2 places) every 6 months (not shown).



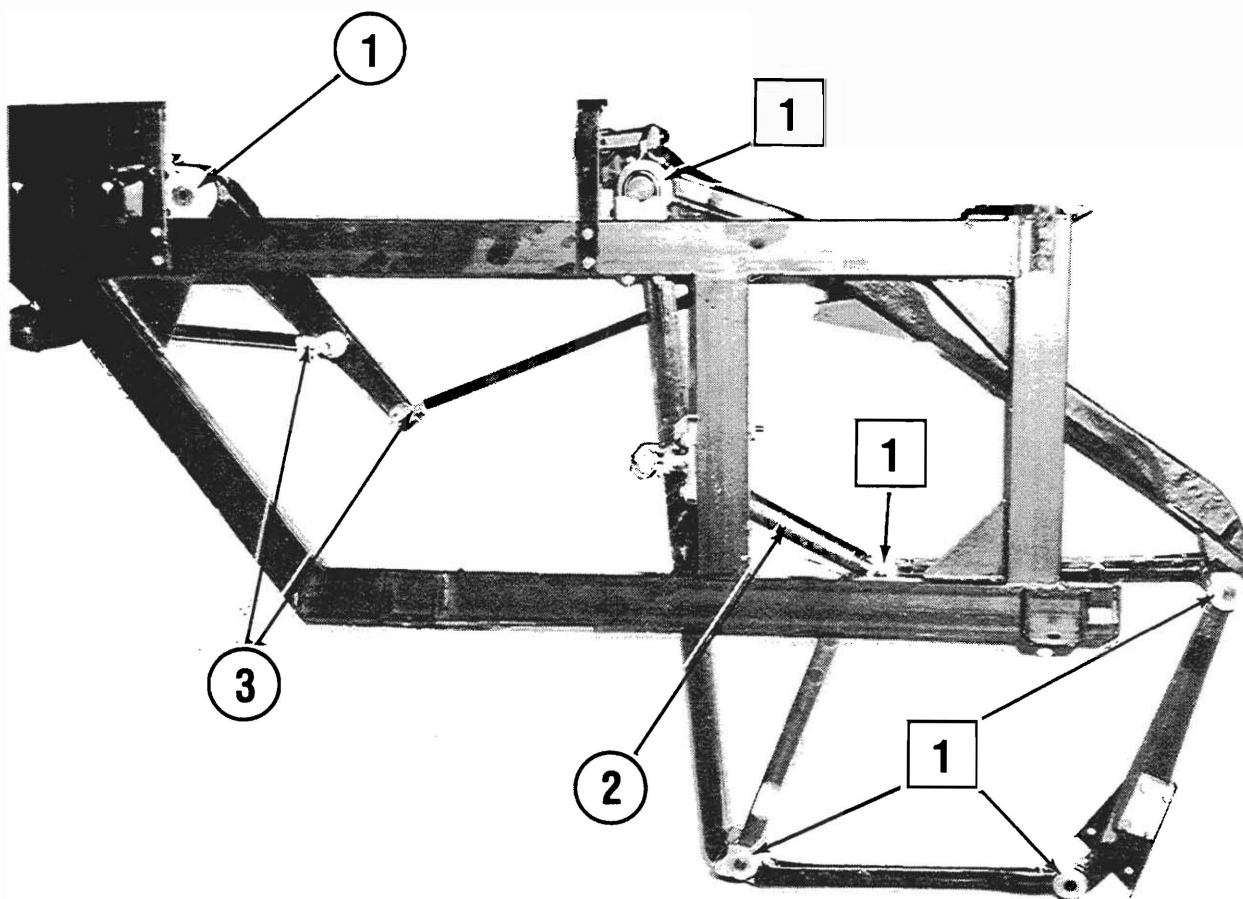
**FRONT END ASSEMBLY
TABLE TRAVEL CAM LEVERS**

- 1** Oilite bearing on pin (1 place) two drops #10 oil every month.
- 2** Oilite bearing at hook pivot (2 places) one drop #10 oil every month.
- 3** Oilite bearings at spot link arm (2 places) 2 drops #10 oil every month.
- 4** Spring mounting plate pivot bolt not shown (2 places) one drop #10 oil every month.
- 1** Grease fitting at cam follower bearings (2 places) apply grease once every month.
- 2** Grease fittings at spot and respot arm (2 places) every 3 months.



**FRONT END ASSEMBLY
TABLE DRIVE ASSEMBLY**

- | | |
|--|--|
| <div style="border: 1px solid black; padding: 2px; display: inline-block; width: 30px; text-align: center; margin-bottom: 10px;">1</div> <div style="border: 1px solid black; padding: 2px; display: inline-block; width: 30px; text-align: center; margin-bottom: 10px;">2</div> <div style="border: 1px solid black; padding: 2px; display: inline-block; width: 30px; text-align: center; margin-bottom: 10px;">3</div> <div style="border: 1px solid black; padding: 2px; display: inline-block; width: 30px; text-align: center; margin-bottom: 10px;">4</div> <div style="border: 1px solid black; padding: 2px; display: inline-block; width: 30px; text-align: center; margin-bottom: 10px;">5</div> <div style="border: 1px solid black; border-radius: 50%; padding: 5px; display: inline-block; width: 30px; text-align: center; margin-bottom: 10px;">1</div> <div style="border: 1px solid black; border-radius: 50%; padding: 5px; display: inline-block; width: 30px; text-align: center; margin-bottom: 10px;">2</div> <div style="border: 1px solid black; border-radius: 50%; padding: 5px; display: inline-block; width: 30px; text-align: center;">3</div> | <p>Oilite bearing in link at roller arm (3 places). One drop #10 oil every month.</p> <p>Oilite bearings in crank housing (2 places). One drop #10 oil every month.</p> <p>Oilite bearing in latch pivot (2 places). One drop #10 oil every month.</p> <p>Oilite bearing in link assembly (4 places). One drop #10 oil every month.</p> <p>Solenoid linkage (6 places) one drop #10 oil every month.</p> <p>Apply grease at crank housing grease fitting for roller bearings once each year.</p> <p>Apply light film of grease to inner surfaces of elongated link slot once every 6 months.</p> <p>Apply a light film of grease at the cam ball every 6 months.</p> |
|--|--|



**FRONT END ASSEMBLY
SWEEP DRIVE LINKAGES**

1

Oilite bearings in sweep pantograph joints (40 places) at 10 joints, 4 drops #10 oil each place every month.

1

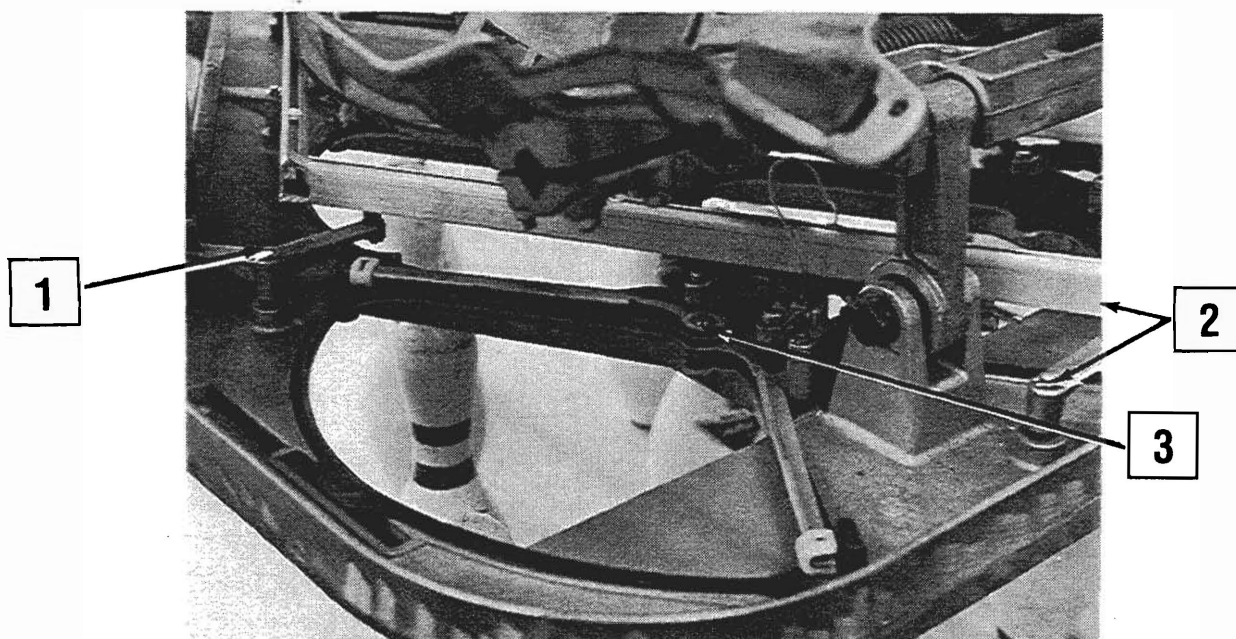
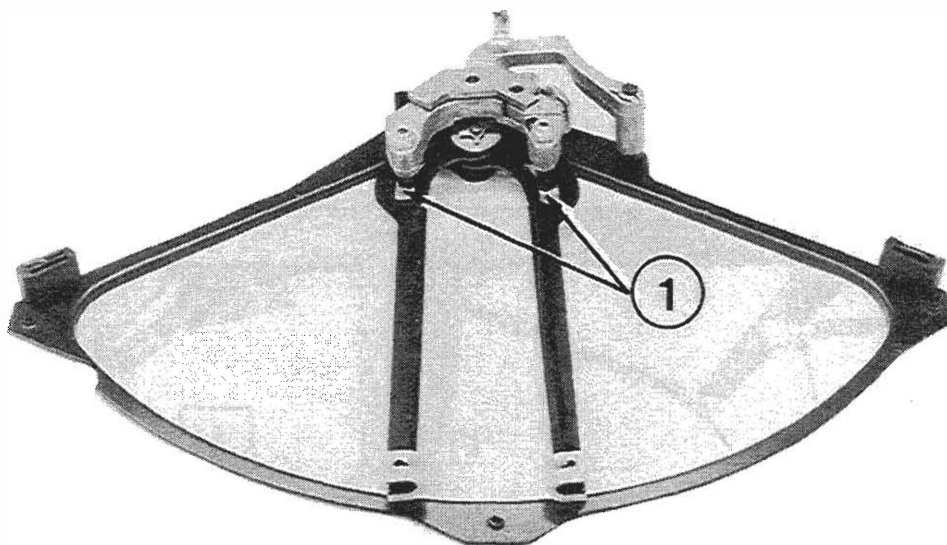
Pillow block bearings, sweep torque tube (2 places) grease every 3 months.

2

Grease in guide tube and rod ends every 3 months.

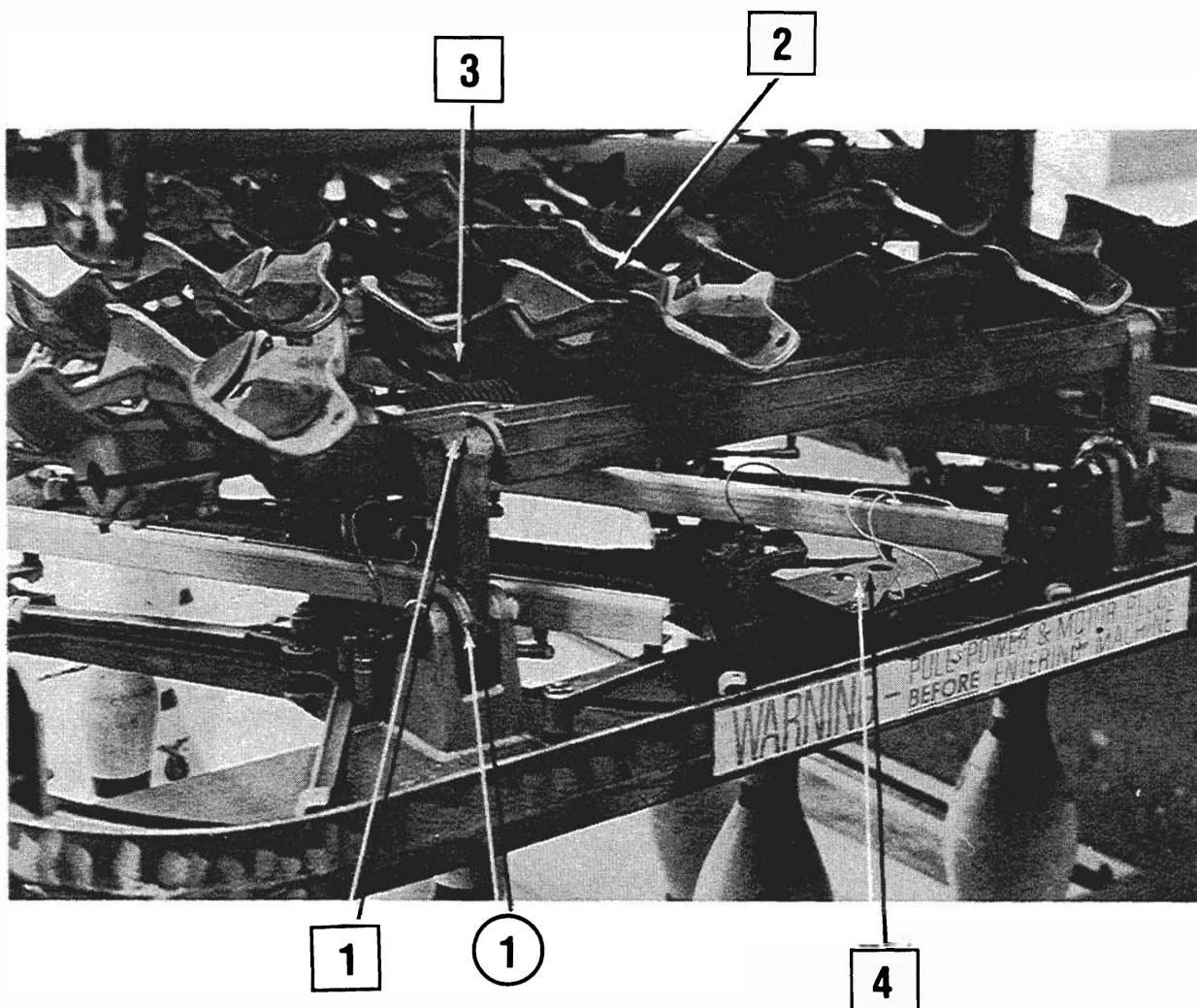
3

Sweep drive connecting rod ends (6 places) grease every 3 months.



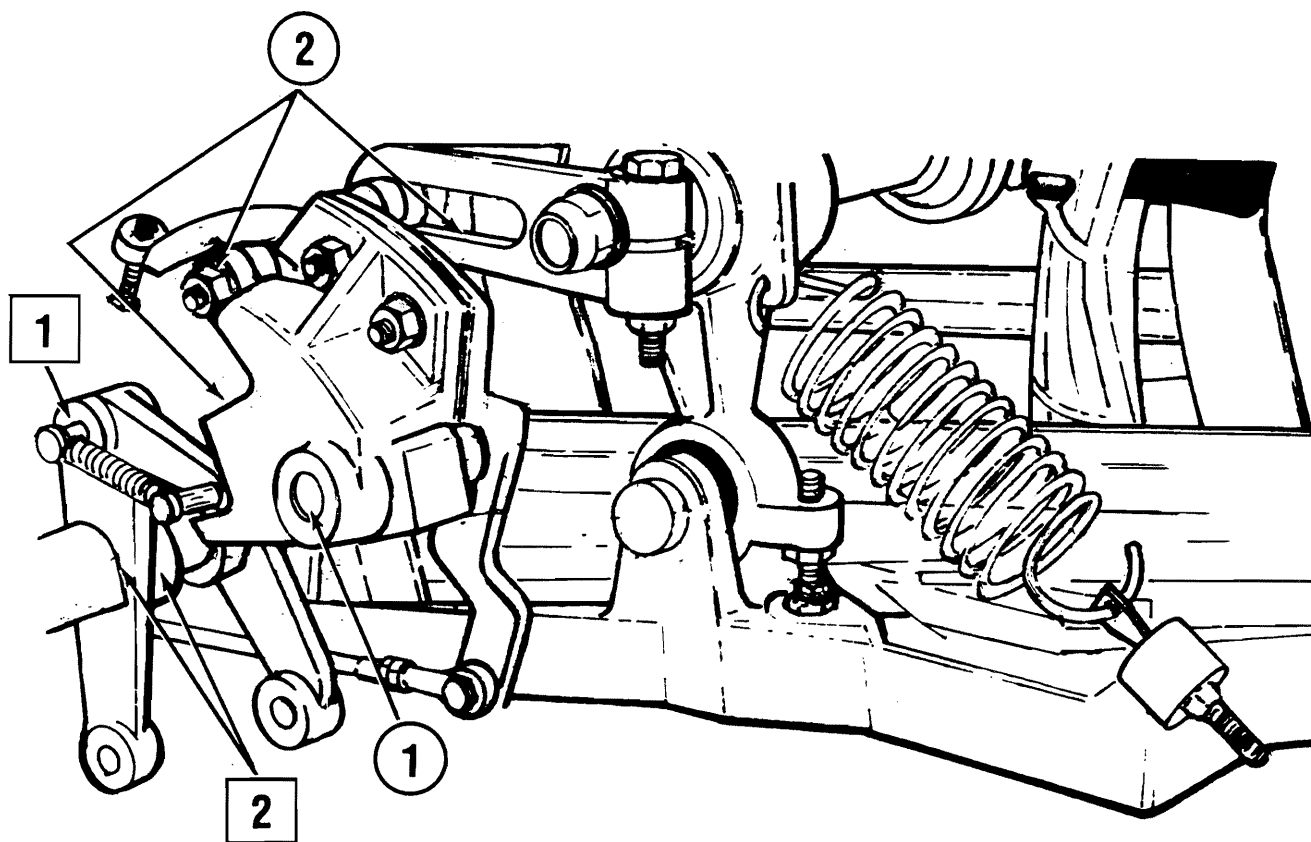
FRONT END ASSEMBLY TABLE RESPOT LINKAGES

- 1** Bearings in (4) respot levers, one illustrated 4 places each lever, 2 drops #10 oil every 3 months.
- 2** Bearings in (3) respot cell levers, one illustrated 3 places each lever, 2 drops #10 oil every 3 months.
- 3** Bearings in respot cell fingers and connecting links 10 cells, 7 places each cell, 2 drops #10 oil every 3 months.
- 1** Stud long, stud short, 2 places, 10 cells light film grease every month.



YOKE ASSEMBLY

- 1** Upper table leg oilite bearings rear legs (4 places) 4 drops #10 oil every 3 months.
- 2** Spotting cup link, oilite bearings (4 places) two drops #10 oil every 3 months.
- 3** On springs of yoke shaft (3 places) 5 drops #10 oil between spring coils every 3 months.
- 4** Respot cell protection switch pivot and roller 4 places, 2 drops #10 oil every 3 months.
- 1** Spherical bearing, front and rear legs (4 places) re-pack with grease once every year.



**FRONT END ASSEMBLY
TABLE SHIFTER MECHANISM**

1

Oilite bearing in finger link (2 places) two drops #10 oil every 3 months.

2

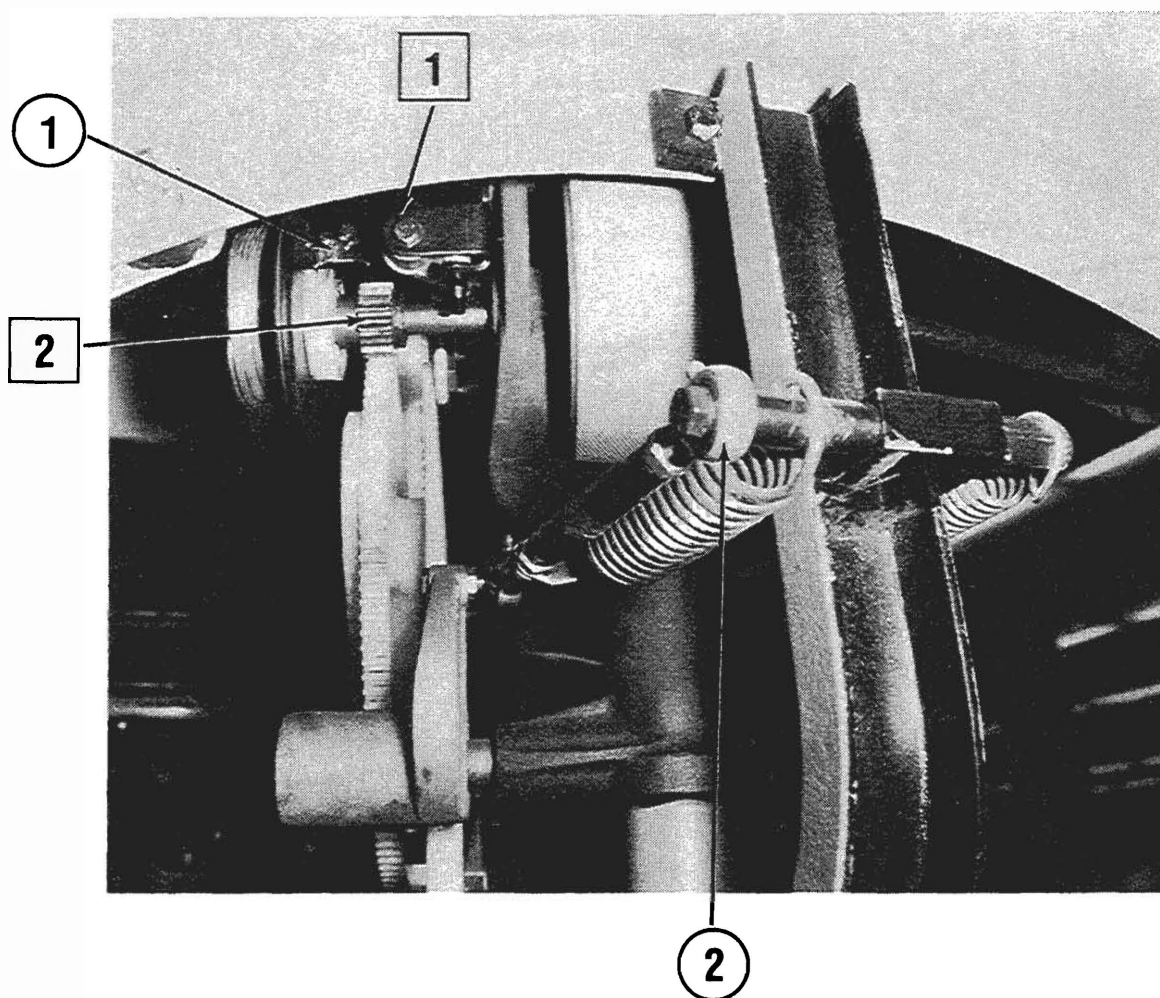
Oilite bearings in shifter link and finger lever assemblies (4 places) 2 drops #10 oil each place every 3 months.

1

Re-pack shifter link needle bearings (2 places) with grease once each year.

2

Apply light film of grease to the surfaces of shifter link and actuator arm (3 places) every 3 months.



DISTRIBUTOR DRIVE

1

Bearings in trip support castings (2 places) 2 drops #10 oil every 3 months.

2

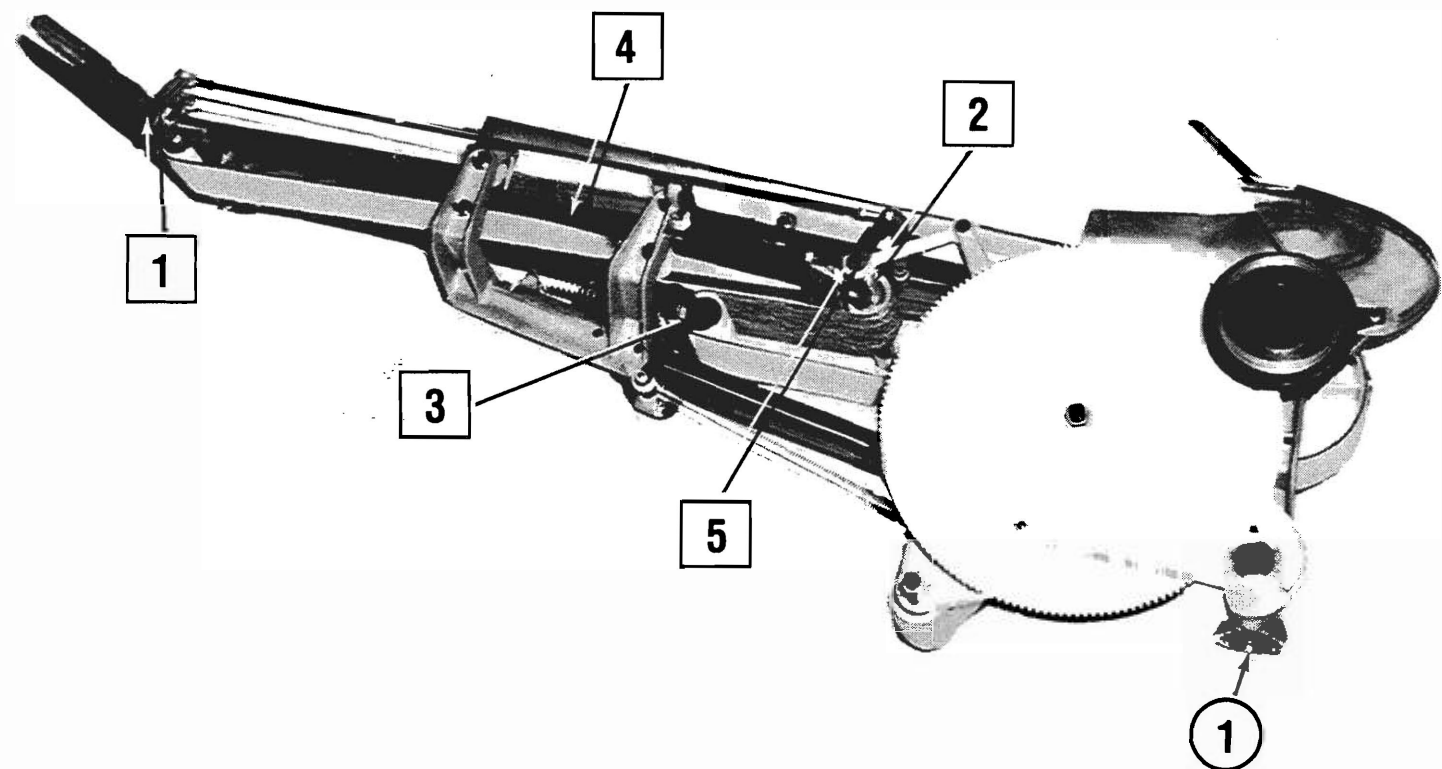
Bearing in drive pinion (1 place) 2 drops of #10 oil every month.

1

Surface of distributor stop blade (2 places) grease every month.

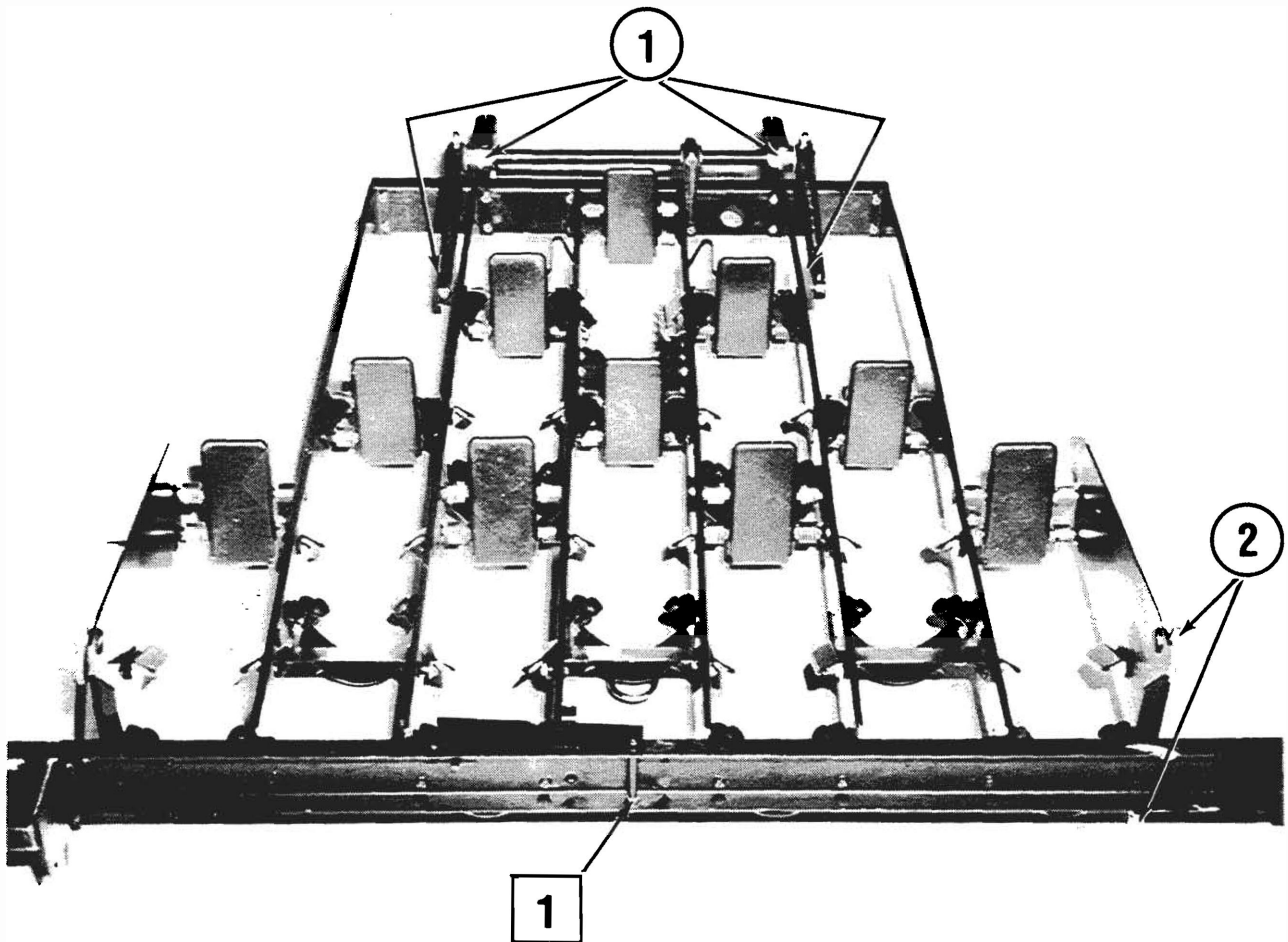
2

On tubing rod ends (2 places) grease every 3 months.



DISTRIBUTOR

- 1** Trip lever, bearings (2 places) 2 drops #10 oil every 3 months.
- 2** Tube housing (2 places) 2 drops #10 oil every 3 months.
- 3** Roller tracking bracket (4 places) 2 drops #10 oil every 3 months.
- 4** On carriage tubes where it contacts nylon bearings (4 places) a light film of #10 oil every 3 months.
- 5** Trip rocker arm shaft (2 places) 2 drops #10 oil every 3 months.
- 1** Bearings in support casting (2 places) grease once a year.



SHUTTLE ASSEMBLY

1

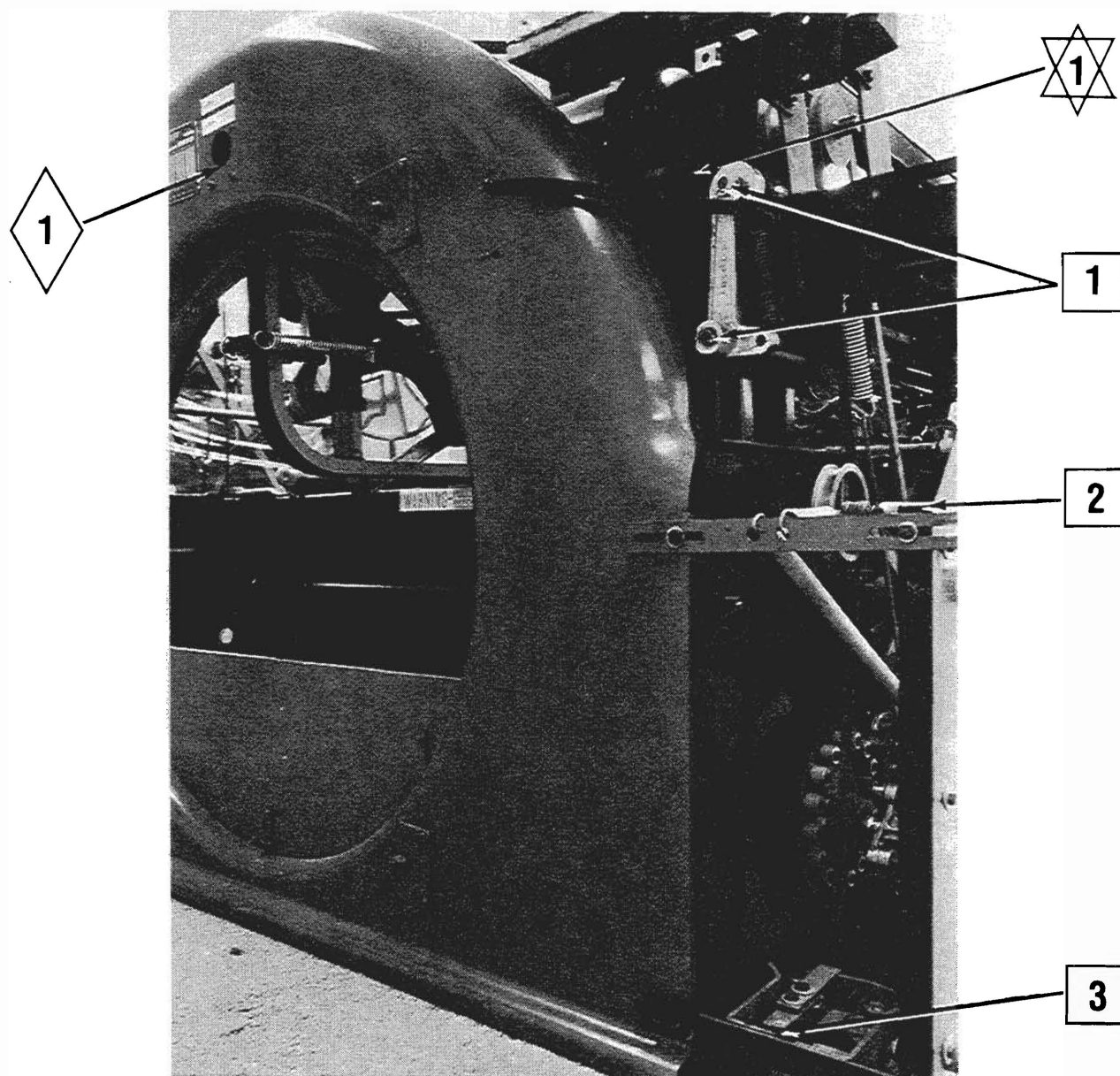
Oilites in bin switch bracket (2 places) 2 drops #10 oil every 6 months.

1

Grease fitting in front levers and shuttle connecting rods (4 places) every 6 months.

2

Grease fittings in rear shuttle connecting rods (4 places) every 6 months.



BACK END ASSEMBLY

1

Oilite bearings in belt tensioner arm (2 places at pivot and one at roller) 4 drops #10 oil every 3 months.

2

Oilite bearings in belt tensioner arms (4 places) 4 drops #10 oil every 3 months.

3

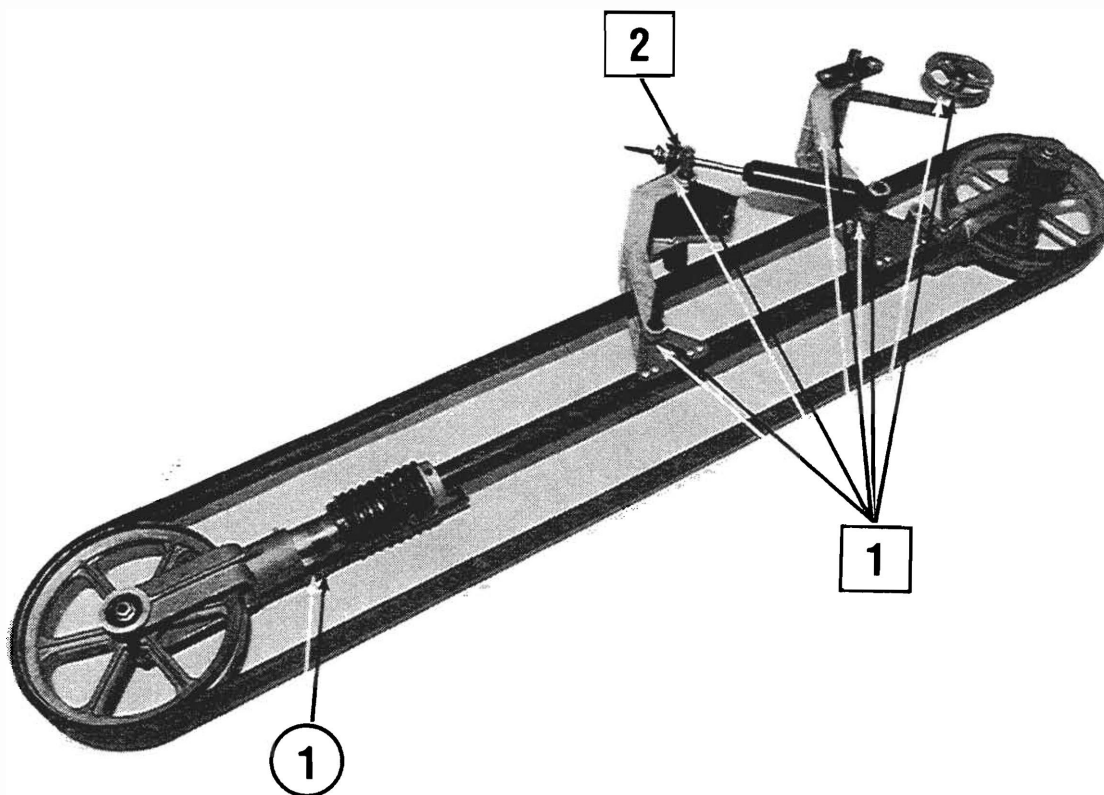
Oilite bearings in rudder arm pivots (2 places) 4 drops #10 oil every 3 months.

1

Check oil level of gear motor at level plug once per year. Fill as required with J-156 (AMF Part No. 715 020 906). Drain and refill every **five years**. (Refer to instructions table—sweep motors page 6.3).

1

Maintain oil level in ring tube oiler reservoir 3/8" below oiler top. Refill as required every year using SAE #80 Way Oil (AMF Part No. 715 021 006).



**BACK END ASSEMBLY
BALL ELEVATOR**

1

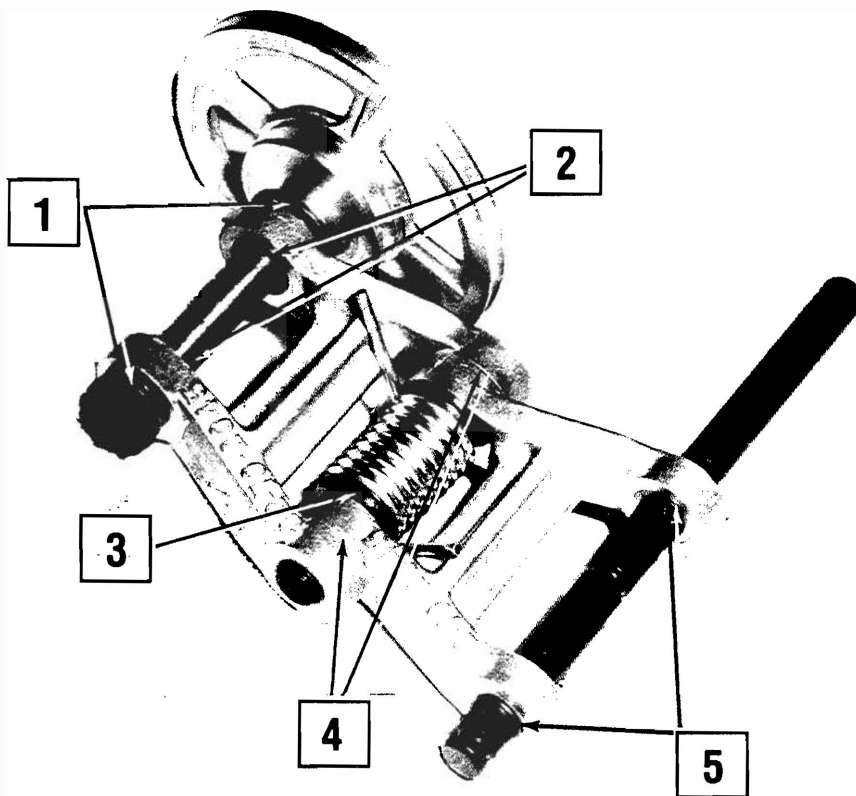
Oilite bearings in links (8 places) 2 drops #10 oil every 3 months.

2

Shock absorber bracket (2 places) 2 drops #10 oil every 3 months.

1

Grease fitting in upper yoke; grease (1 place) every 3 months.



BELT TENSIONER

1

Oilite thrust bearings at pulley shaft (2 places) 2 drops #10 oil every 3 months.

2

Oilite bearings at hanger arm pulley shaft (2 places) 4 drops #10 oil every 3 months.

3

Oilite bearing spring guide (1 place) 2 drops #10 oil every 3 months.

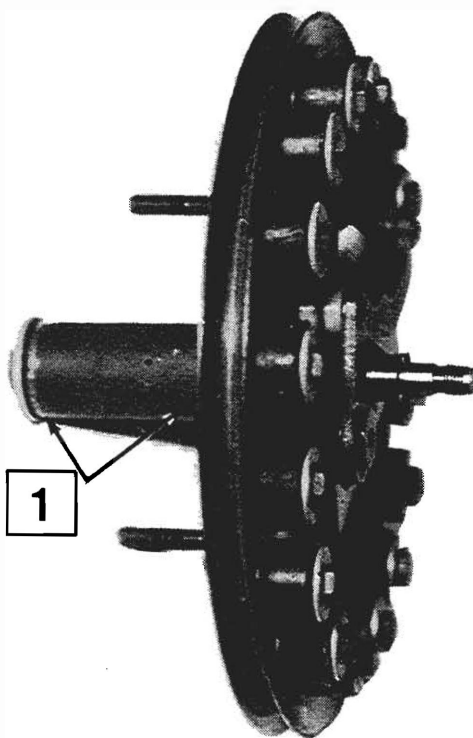
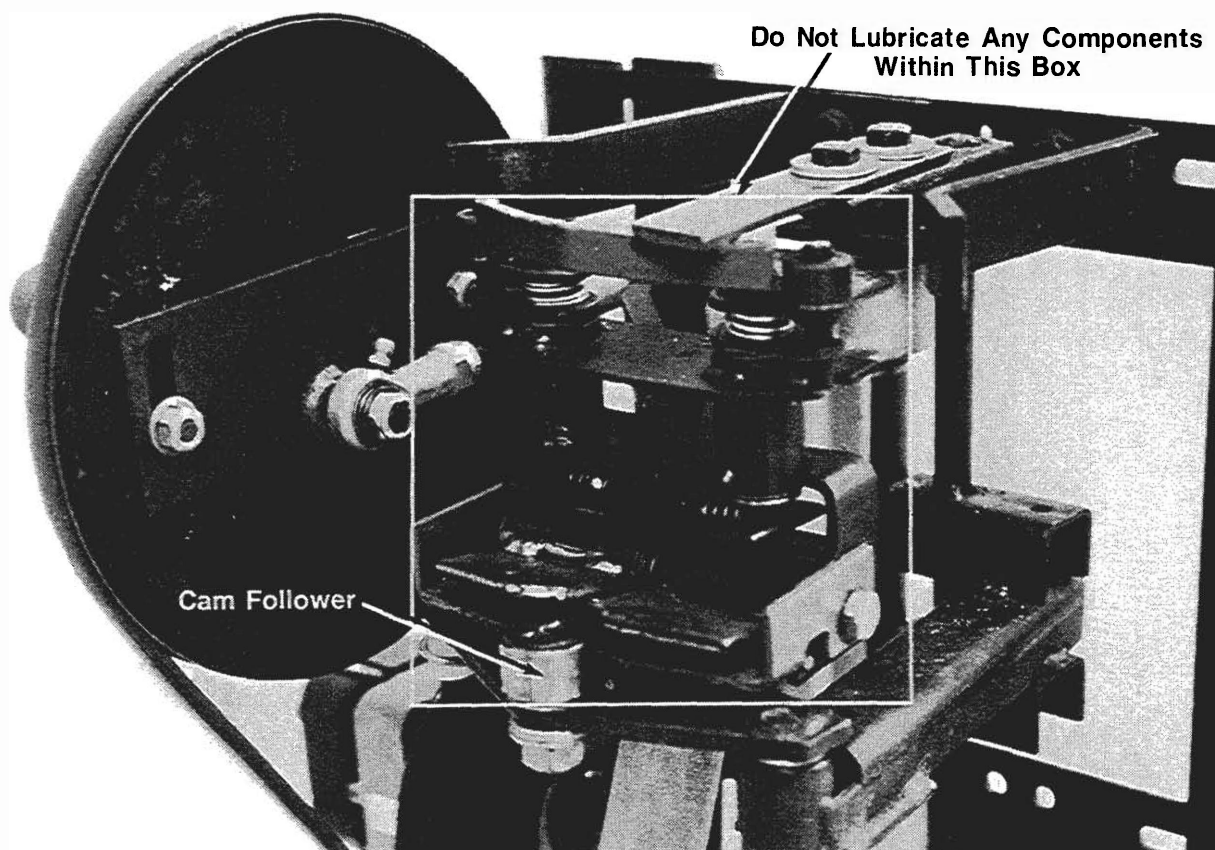
4

Oilite bearings at hanger arm spring shaft (2 places) 2 drops #10 oil every 3 months.

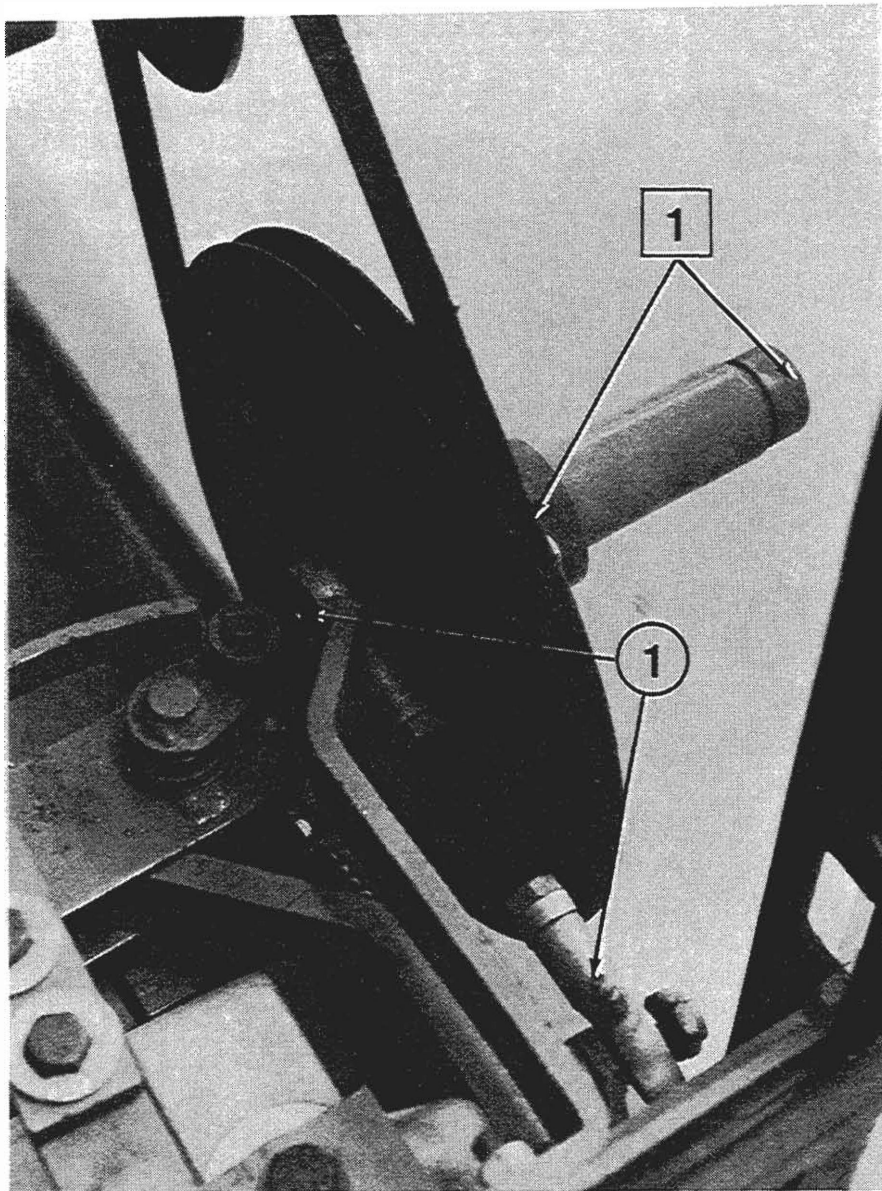
5

Oilite bearings at pivot shaft (2 places) 2 drops #10 oil every 3 months.

LIGHT BALL SENSOR

**1**

Flange bearings on ratchet wheel shaft (2 places) 2 drops #10 oil every 3 months.



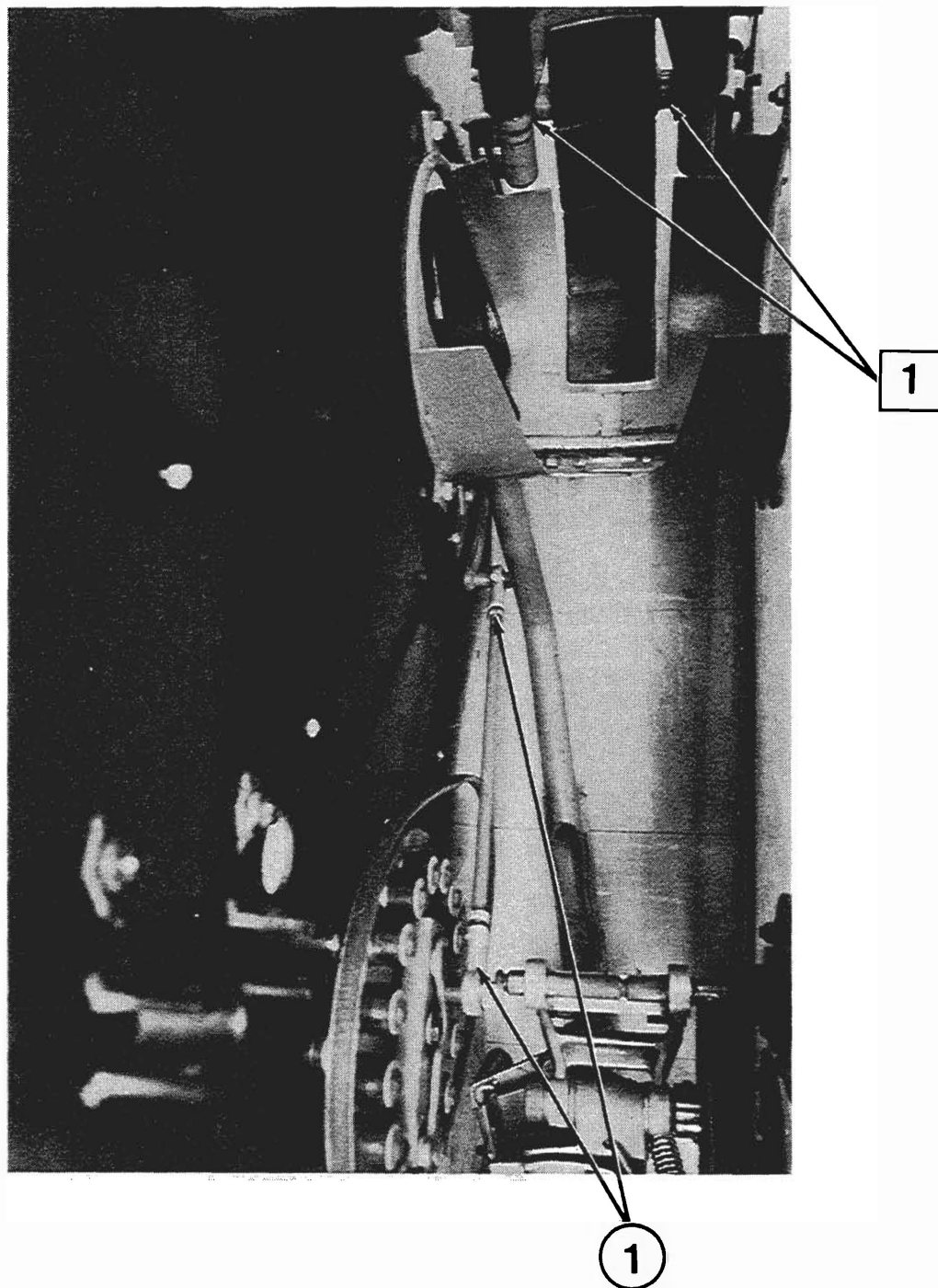
RUDDER DRIVE ASSEMBLY

1

Oilite bearings in shaft tube (2 places) 2 drops #10 oil every 3 months.

1

Apply grease at both rod ends every 3 months.



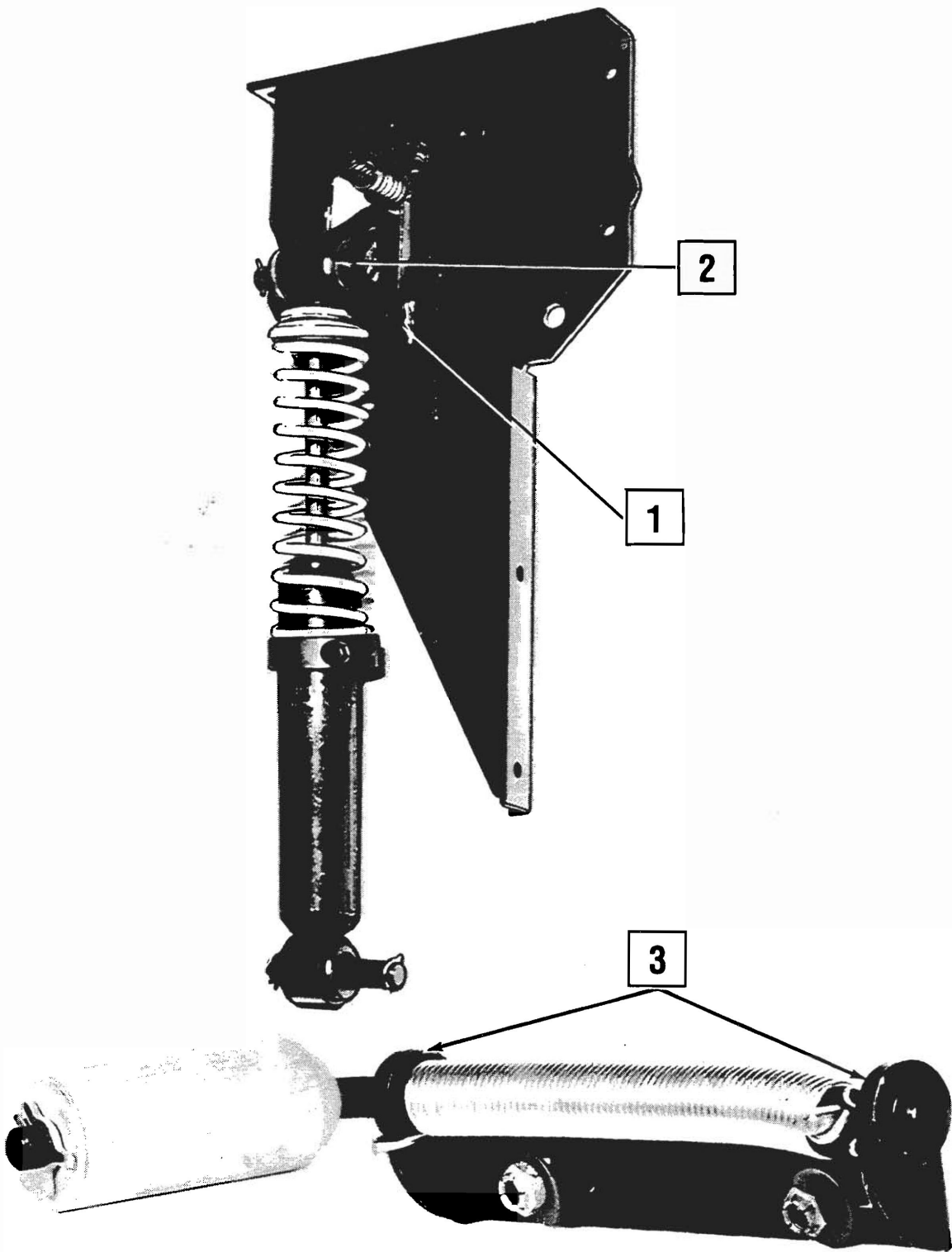
TRACK RAIL ASSEMBLY

1

Flange bearings on lift arm (2 places) #10 oil every 3 months.

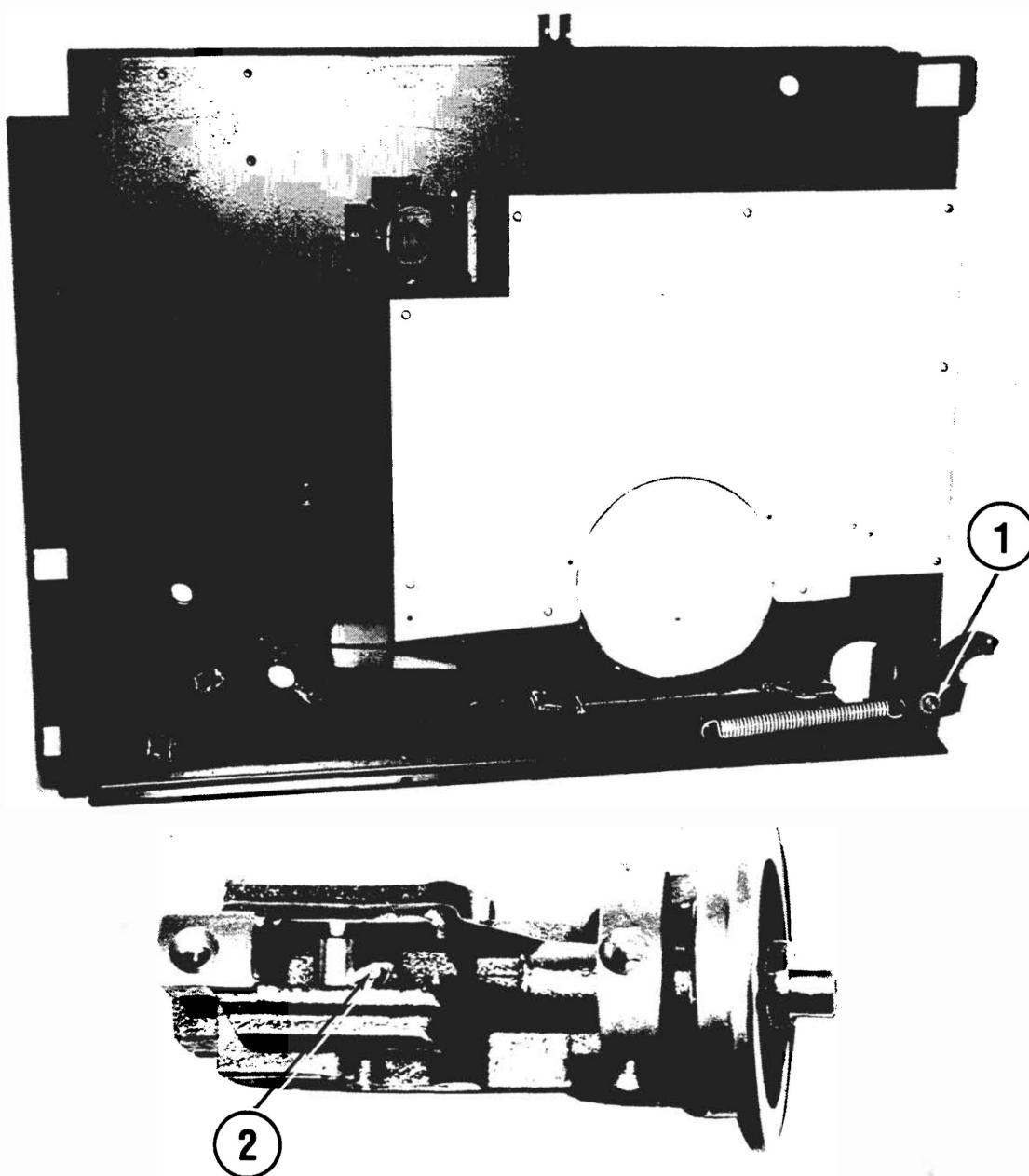
1

Apply grease at fittings (2 places) every 3 months.



BACK END ASSEMBLY

- | | |
|----------|--|
| 1 | Pivots of start switch lever and shock mounting bracket (2 places) 2 drops #10 oil every 3 months. |
| 2 | Pivot pin of shock absorber (2 places) 2 drops #10 oil every 3 months. |
| 3 | Oilite bearings in pin ejector assembly (2 places) 2 drops #10 oil every 3 months. |



BACK END ASSEMBLY

- 1** Grease fitting in front roller support brackets (2 places) left and right sides apply grease every 6 months.
- 2** Remove the front roller, apply grease through access hole to grease fitting (2 places) every six months.

PINSPOTTER LUBRICATION SCHEDULE

	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
MOTORS	X											
CAM LEVERS/PIVOTS	X			X			X			X		
SPOT/RESPOT CAMS	X	X	X	X	X	X	X	X	X	X	X	X
TABLE SPHERICAL BUSHING	X			X			X			X		
TABLE CONN. RODS/BUSHINGS	X			X			X			X		
OVER TRAVEL LEVER	X						X					
OILITES IN SPOT/RESPOT LINK	X	X	X	X	X	X	X	X	X	X	X	X
SPOT/RESPOT ARM FITTINGS	X			X			X			X		
SOLENOID LINKAGE	X	X	X	X	X	X	X	X	X	X	X	X
TABLE DRIVE	X											
SWEEP PANTOGRAPH LINKS	X	X	X	X	X	X	X	X	X	X	X	X
SWEEP CONN. RODS/BEARINGS	X			X			X			X		
RESPOT CELLS	X			X			X			X		
YOKE ASSEMBLY	X			X			X			X		
YOKE SPHERICAL BEARINGS	X											
SHIFTER LINK	X											
RESPOT SHIFTER MECHANISM	X			X			X			X		
DISTRIBUTOR PINION/STOP BLADE	X	X	X	X	X	X	X	X	X	X	X	X
DISTRIBUTOR	X			X			X			X		
SHUTTLE ASSEMBLY	X											
BELT TENSIONER OILITES	X			X			X			X		
RING TUBE OILER	X											
BALL ELEVATOR	X			X			X			X		
RUDDER DRIVE	X			X			X			X		
TRACK RAIL ASSEMBLY	X			X			X			X		
START SWITCH PIVOTS	X			X			X			X		
PIN EJECTOR ASSEMBLY	X			X			X			X		
FRONT ROLLERS & SUPPORTS	X						X					